

Suggested Solutions to 2013 H2 Physics G.C.E. A Level Examination Paper 3

Q.	Solution
1(a)	Gravitational potential at a point in space is defined as the work done per unit mass by an external force in bringing a mass from infinity to the point without any change in kinetic energy.
1(b)	Gain in KE = Loss in GPE $\frac{1}{2}mv^2 - 0 = 0 - \left(-\frac{GMm}{R}\right)$ $v = \sqrt{\frac{2GM}{R}}$ $= \sqrt{\frac{2(6.67 \times 10^{-11})(6.4 \times 10^{23})}{3.4 \times 10^6}}$ $= 5011 \text{ m s}^{-1}$ $= 5.0 \times 10^3 \text{ m s}^{-1} \text{ (2 s.f.)}$
1(c)(i)	$c_{\text{rms}} = \sqrt{\frac{3RT}{M_m}}$ $5011 = \sqrt{\frac{3(8.31)T}{0.004}}$ $T = 4.0 \times 10^3 \text{ K}$
1(c)(ii)	Yes helium-4 gas is found on the surface of Mars. Since the surface temperature of Mars is much lower than $4.0 \times 10^3 \text{ K}$, the root-mean-square speed of helium-4 gas on Mars is much lower than $5.0 \times 10^3 \text{ m s}^{-1}$. Hence they would not be able to escape from Mars.
2(a)(i)	For $R = 4.0 \Omega$, $P_R = 9.0 \text{ W}$